PharmAchieve

ONYCHOMYCOSIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

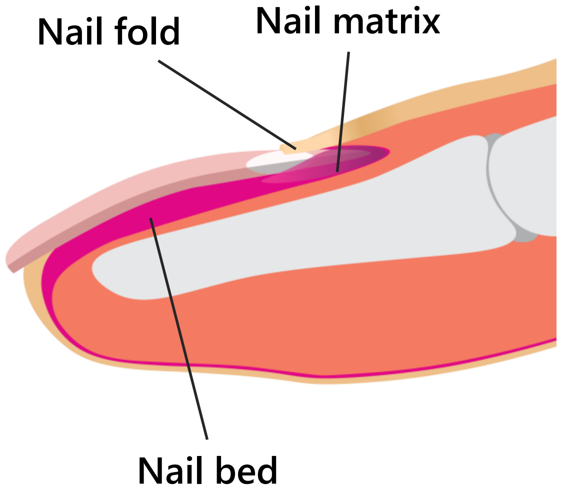
LEGEND

AIDS	Acquired Immunodeficiency Syndrome
BID	Bis In Die (Twice Daily)
BPH	Benign Prostatic Hyperplasia
CKD	Chronic Kidney Disease
COPD	Chronic Obstructive Pulmonary Disease
DLSO	Distal & Lateral Subungual Onychomycosis
DM	Diabetes Mellitus
H	Hours
LFTs	Liver Function Tests
OM	Onychomycosis
PO	Per Os (By Mouth)
PRN	Pro Re Nata (As Needed)
PSO	Proximal Subungual Onychomycosis
PVD	Peripheral Vascular Disease
Q	Quaque (Every)
SWO	Superficial White Onychomycosis
T2DM	Type 2 Diabetes Mellitus
TDO	Total Dystrophic Onychomycosis

PATHOPHYSIOLOGY

FUNGAL INFECTION OF THE NAIL

- Fungal invasion of nail matrix that significantly changes nail appearance
- Organisms responsible:
  - Dermatophytes: *Trichophyton rubrum* (most common)
    - Dermatophyte OM is also known as "tinea unguium"
  - Yeast: *Candida* species
  - Non-dermatophytes (rare)
- Chronic condition that can progress to other nails or skin areas
- Rarely spontaneously resolves
- In patients with diabetes or who are immunocompromised, infected area can serve as a reservoir leading to reinfection or secondary bacterial infections
- Classified into 5 different subtypes



DIFFERENTIATION

FYI

SUBTYPES OF ONYCHOMYCOSIS

Distal lateral subungual onychomycosis (DLSO)	Most common with intercurrent tinea pedis Thickening, discoloration (white-yellow-brown) of distal and/or lateral edge Crumbling yellow debris under nail edge (subungual hyperkeratosis)
Endonyx onychomycosis	Rare; involves only the interior of the nail plate (no nail bed involvement) Milky white patches (discolouration) and splitting of distal nail plate No subungual hyperkeratosis
Proximal subungual onychomycosis (PSO)	Uncommon; seen in severely immunocompromised patients (AIDS marker) Crumbling and whitening of proximal area beneath nail bed
Total dystrophic onychomycosis (TDO)	End-stage DLSO or PSO; complete destruction of nail plate Nail plate and matrix are infected; thickened nail bed with keratotic debris Nail appears ridged and opaque with a yellow-brown colour
Superficial white onychomycosis (SWO)	Most common in children Nail plate is infected from above with no involvement of the nail bed Nail plate surface appears to have dull white and powdery patches

DIFFERENTIATION

FYI

YEAST ONYCHOMYCOSIS

Chronic paronychia	Common in occupations with frequent hand immersion in water Erythematous swelling of the nail fold (slow onset) Discolouration of the nail (brown or bluish-black)
Distal nail candidal infection	Common with some forms of peripheral vascular disease Presence of subungual hyperkeratosis that is yellowish-grey in appearance Leads to onycholysis (painless separation of the nail from the nail bed)
Chronic mucocutaneous candidiasis	Often seen in immunocompromised patients Increased thickening of the nails can result in a candida granuloma Causes brittle nails and can progress to involve mucous membranes
Secondary candidal onychomycosis	Commonly seen in nails previously damaged from trauma or disease Worsening of pre-existing dystrophic nail changes Clinical presentation varies

RISK FACTORS

Dermatologic diseases	Tinea pedis, psoriasis, hyperhidrosis
Comorbidities	Diabetes, immunosuppression, venous insufficiency, malignancy, peripheral artery disease, obesity, inflammatory bowel disease
Exposures	Trauma, poor nail grooming, sports and fitness activities, occupation, smoking, occlusive shoes
Other	Advancing age, contact with infected household members, genetic predisposition

Candida albicans onychomycosis can occur with frequent exposure of the hands to moisture, as seen in occupations such as bartenders or housekeepers

Rarely, severe candidal onychomycosis is a manifestation of chronic mucocutaneous candidiasis

DIFFERENTIAL DIAGNOSIS

Diagnosis	Clinical Presentation
Bacterial paronychia	Fast onset of painful swollen nail fold or cuticle and can have vesicles filled with pus
Eczema	Transverse ridges with affected nail folds
Lichen planus	Longitudinal ridges with nail atrophy
Medications (e.g. chemotherapy)	Onycholysis
Melanoma	Dark vertical bands
Nail psoriasis	Symmetrical nail pitting with yellow-gray or silvery white nails
Onychogryphosis	Thick, claw-like nails (seen in elderly)
Yellow-nail syndrome	Slow nail growth with yellow-green curved appearance

GOALS OF THERAPY

- ✓ Reduce or eradicate infection
- ✓ Improve nail appearance
- ✓ Prevent transmission of infection
- ✓ Prevent recurrence of infection
- ✓ Prevent complications including secondary bacterial infections

NON-PHARMACOLOGICAL THERAPY

PREVENTION

1. Wear footwear and socks that minimizes humidity
2. Dry feet thoroughly after washing
3. Keep nails clean and cut short straight across
4. Do not share nail clippers or footwear
5. Wear rubber gloves if hands often immersed
6. Control chronic comorbidities (DM, PVD)
7. Avoid being barefoot in shared public places
8. Give feet a break from shoes
9. Examine nails regularly

TREATMENT

- Apply emollients on cracked skin to reduce further spread of infection
- Educate patients on nail trimming and debridement (especially if applying a topical pharmacological agent)

PHARMACOLOGICAL THERAPY OVERVIEW

ALGORITHM

Treatment of dermatophyte OM*

- **>60% nail involvement** → Oral terbinafine
- **20-60% nail involvement** → Topical efinaconazole ± oral terbinafine (>3 nails)
- **<20% nail involvement** → Topical efinaconazole
  - May be combined with:
    - Selective debridement
    - Oral terbinafine

Evaluation strategies:

- Evaluate clinical/microbiological outcomes
- Address treatment modifiers for OM
- Explore barriers to adherence

Maintenance (reduce the risk of reinfection):

- Lifestyle modifications
- Regular follow-up
- Periodic self-examination
- Treat nails previously infected or reinfected with topical efinaconazole

* Supportive evidence is based on toenail OM

PHARMACOLOGICAL THERAPY

TOPICAL AGENTS

Class	Dosing and Duration	Cure Rates - FYI	Comments
Ciclopirox 8%	Apply once daily until nail clearance or up to 48 weeks	Mycotic: 33% Clinical: 7%	• Used in SWO or mild/moderate DLSO • Clean nails weekly with isopropyl alcohol • Flammable
Efinaconazole 10% (JUBLIA®)	Apply 1 drop (2 drops for big toe) nightly for 48 weeks	Mycotic: 54-61% Clinical: 14-47%	• Used in mild/moderate DLSO • Flammable
Propylene glycol 66.4%/urea 20%/lactic acid 10% (Emtrix®)	Apply once daily to the tip and under the nail for 24 weeks	Mycotic: 27.2%	• Used in mild DLSO • Keratolytic and hydrating effects improve appearance • Wash from unaffected skin thoroughly

PHARMACOLOGICAL THERAPY

ORAL AGENTS

Class	Dosing and Duration	Cure Rates - FYI	Comments
Terbinafine	Adults: 250 mg daily Fingernails: 6-12 weeks Toenails: 12-24 weeks	Mycotic: 70% Clinical: 38%	• Effective for dermatophyte OM • Monitor LFTs • CYP2D6 inhibitor (least drug interactions amongst oral agents)
Itraconazole (Sporanox®)	*Pulsed*: 200 mg BID Fingernails: 1 week/month x 2 Toenails: 1 week/month x 3 *Continuous*: 200 mg daily Fingernails: 6 weeks Toenails: 12 weeks	Mycotic: 54% Clinical: 14%	• Preferred treatment for non-dermatophyte OM • Monitor LFTs • Less side effects with pulsed therapy • CYP3A4 inhibitor
Fluconazole (off-label)	Adults: 150-300 mg weekly Fingernails: 12-16 weeks Toenails: 18-26 weeks	Mycotic: 55-59% Clinical: 20%	• Least cost effective • Many drug interactions • QTc prolonging agent • CYP2C19, CYP2C9, and CYP3A4 inhibitor

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Infection	Eradicate	6-48 weeks (depending on agent used)
Nail Appearance	Improve	6-48 weeks (depending on agent used)
Secondary bacterial infection	Prevent	Duration of therapy
Recurrence of infection	Prevent	Ongoing

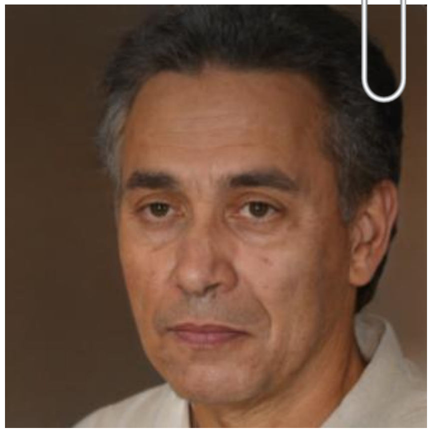
SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Local skin irritation	Minimal to none	Duration of therapy
Elevation of LFTs	None	Baseline, midway (at 4-6 weeks)
QTc prolongation	None	Duration of therapy

TAKEAWAY

- Onychomycosis can cause disfigurement of the nail, pain, and may increase soft tissue bacterial infection in patients who are immunocompromised or have diabetes mellitus
- Dermatophytes (e.g. *Trichophyton rubrum*) are the most common causes of toenail onychomycosis; yeasts or non-dermatophyte moulds may cause onychomycosis
- There are 5 subtypes; DLSO is the most common clinical subtype
- Prevention strategies: wear proper footwear and socks, keep feet clean and dry, take breaks from shoes, keep nails clean and trimmed, avoid being barefoot in public spaces, and examine nails regularly
- Emollients may be applied to cracked skin to reduce further spread of infection
- Topical agents are preferred for mild or moderate cases of onychomycosis
    - Efinaconazole, ciclopirox, or propylene glycol/urea/lactic acid
- Oral agents are preferred for severe cases of onychomycosis (>60% nail involvement)
    - Terbinafine, itraconazole or fluconazole (off-label)

CASE

MX is a 68-year-old male who presents to the clinic with complaints of nail changes over the past two months. He explains that his big toenail on his right foot has gradually turned brown in color, and he has noticed crumbling yellow debris under the edges of the nail. His past medical history is significant for type 2 diabetes mellitus (T2DM), chronic kidney disease (CKD), hypertension, chronic obstructive pulmonary disease (COPD), and benign prostatic hyperplasia (BPH). His current medications include metformin 1000 mg PO BID, empagliflozin 25 mg PO daily, linagliptin 5 mg PO daily, candesartan 16 mg PO daily, salbutamol 200 mcg 1 inhalation Q4-6H PRN, Trelegy Ellipta® (fluticasone furoate 200 mcg/umeclidinium 62.5 mcg/vilanterol 25 mcg) 1 inhalation once daily, alfuzosin 10 mg PO daily, and dutasteride 0.5 mg PO daily. On physical examination, the physician determined that 25% of the distal nail tip was affected, with minimal thickening confined to the nail surface. MX's physician approaches you for your advice regarding therapeutic choices.



1. Which subtype of onychomycosis is MX most likely experiencing?
   a) Proximal subungual onychomycosis
   b) Secondary candidal onychomycosis
   c) Distal lateral subungual onychomycosis
   d) Distal nail candidal infection

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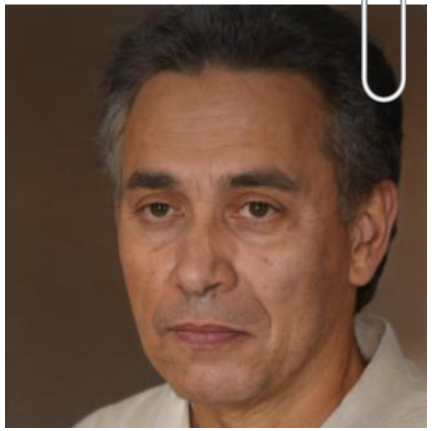
2. Which of the following interventions is the most appropriate to recommend at this time?
   a) Initiate topical efinaconazole 10% nightly and oral terbinafine 250 mg PO daily
   b) Initiate fluconazole 150 mg PO weekly
   c) Initiate terbinafine 250 mg PO daily
   d) Initiate topical efinaconazole 10% nightly



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3. What is the recommended duration of therapy?
   a) 6 weeks
   b) 12 weeks
   c) 24 weeks
   d) 48 weeks

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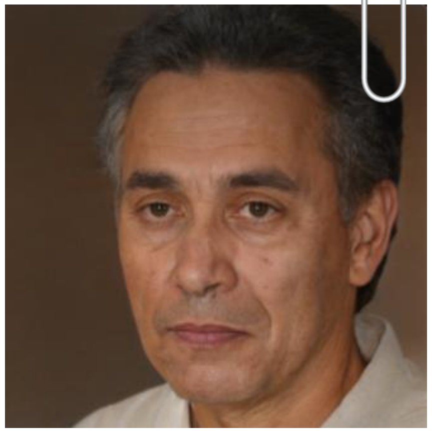
4. MX's episode of onychomycosis resolved after completing the course of therapy. However, two years later, MX returns to the clinic with the same complaint. His physician determines that 80% of the same nail has been affected. Which of the following regimens is most appropriate to recommend for MX?
   a) Terbinafine 250 mg PO daily for 20 weeks
   b) Itraconazole 200 mg PO BID for 1 week per month, for 3 months
   c) Fluconazole 150 mg PO weekly for 24 weeks
   d) Efinaconazole 10% applied nightly (48 weeks) and terbinafine 250 mg PO daily (24 weeks)



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CHANGE LOG

March 2025
- Slide 6: Changed endonyx subungual onychomycosis to endonyx onychomycosis
- Slide 9: Changed criteria for terbinafine from ≥60% to >60% and other symbol for less severe disease also amended. In line with Canadian references and rxtx

July 2025
- Minor formatting changes made, legend updated

September 2025
- Formatting, grammar, and spelling changes made throughout
- Slide 3: Updated legend
- Slide 7: Added risk factors slide
- Slides 9, 10, 11: Rearrangement of slides: Goals of Therapy, Non-Pharmacological Therapy, and Pharmacological Therapy algorithm
- Slide 10: Added prevention strategies
- Slide 11: Updated algorithm
- Slides 12, 13: Added 'FYI' to cure rates. Added brand names, and CYP interactions
- Slide 14: Added monitoring parameters slide
- Slide 15: Added takeaway slide
- Slides 16-19: Added case study and questions
- Slide 20: Updated references

February 2026
- Slide 2: Updated NAPRA competencies
- No content changes